

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 43

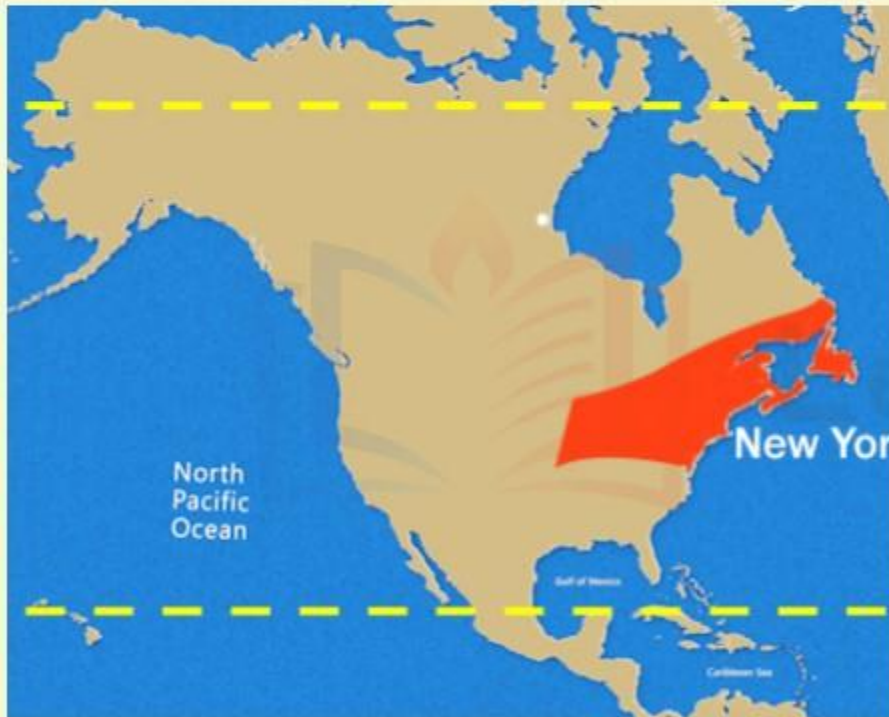
WORLD CLIMATIC TYPES
THE LAURENTIAN CLIMATE

WORLD CLIMATIC TYPES

8

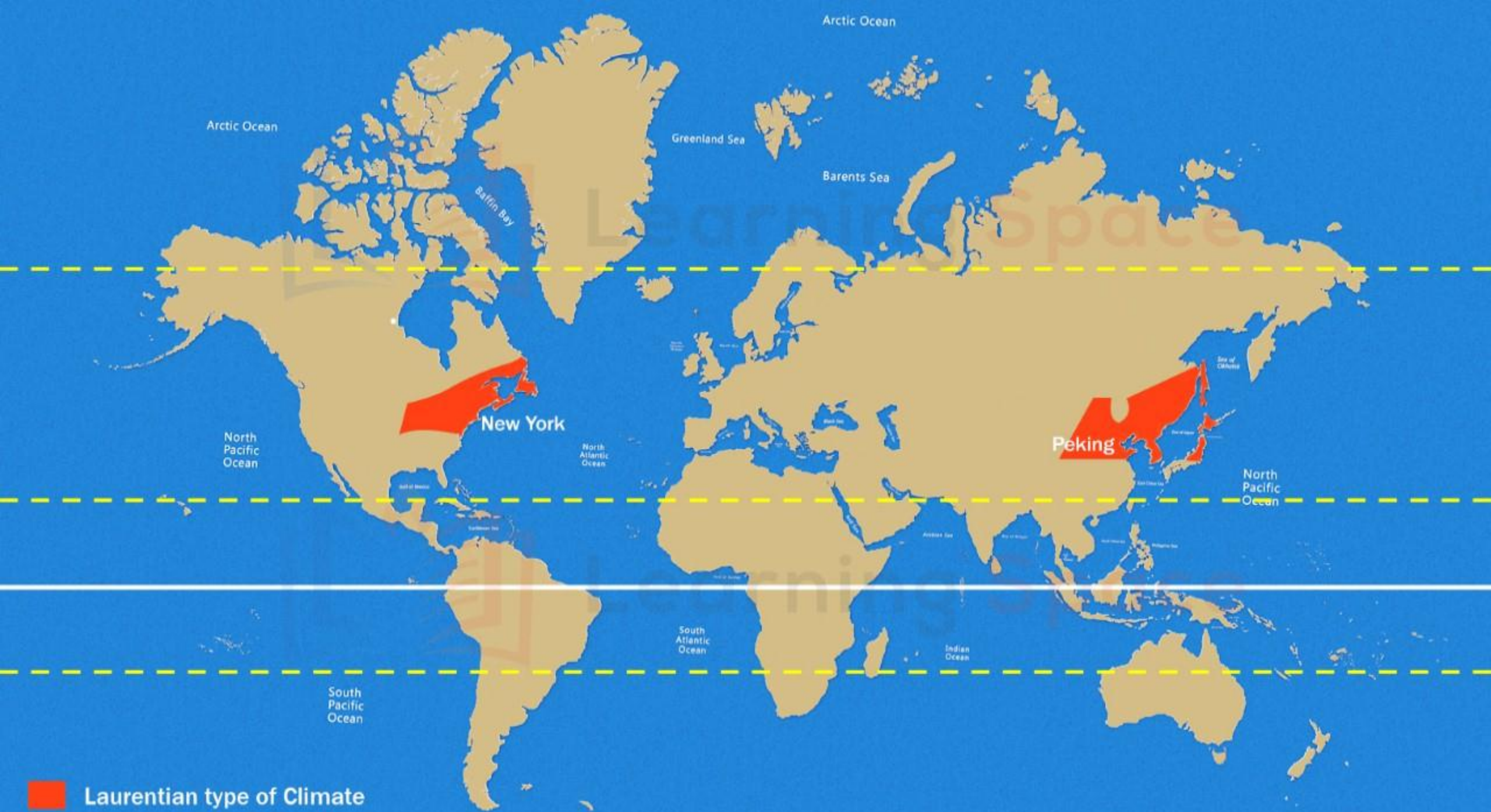
THE COOL TEMPERATE EASTERN MARGIN (LAURENTIAN TYPE) CLIMATE

DISTRIBUTION



- 1 The Laurentian Climate is an intermediate type of climate between the British and Siberian type of climate.
- 2 It has features of both the maritime and the continental climates.
- 3 The Laurentian type of climate is found only in two regions.
- 4 One is north-eastern North America, including eastern Canada, north-east USA. (i.e maritime provinces and the New England States, and Newfoundland). This may be referred to as the North American region.

COLD ZONE	<u>Arctic or Polar Type</u>			90°
	Tundra vegetation, mosses, lichens			
COOL TEMPERATE ZONE				65° N
	Western Margin <u>British type</u> Deciduous forests	Central Continent <u>Siberian type</u> Evergreen, coniferous forests	Eastern Margin <u>Laurentian type</u> Mixed forests	
WARM TEMPERATE ZONE				45° N
	<u>Mediterranean type</u> Mediterranean Forests & shrubs	<u>Steppe type</u> Steppe temperate grassland	<u>China type</u> Warm wet forests	
HOT ZONE				30° N
	<u>Hot desert</u> Desert vegetation	<u>Sudan type</u> Savanna, tropical grassland	<u>Monsoonal type</u> Monsoon forests	
EQUATORIAL ZONE				10° N
	<u>Hot Wet Equatorial Climate</u> Equatorial rain forests			0°



WORLD CLIMATIC TYPES

8

THE COOL TEMPERATE EASTERN MARGIN (LAURENTIAN TYPE) CLIMATE

DISTRIBUTION



5

The other region is the eastern coastlands of Asia, including eastern Siberia, North China, Manchuria, Korea and northern Japan. It may be referred to as **the Asiatic region**.

6

In the southern hemisphere, this climatic type is absent because only a small section of the southern **continents** extends south of the latitude of 40° S.

8

THE COOL TEMPERATE EASTERN MARGIN (LAURENTIAN TYPE) CLIMATE

CLIMATE

- 1 The Laurentian type of climate has cold, dry, winters and warm, wet summers.
- 2 Winter temperature, may be well below freezing-point and snow falls to quite a depth.
- 3 Summers are as warm as the tropics (21° - 27°C) and if it were not for the cooling effects of the offshore cold currents from the Arctic the summer might be even hotter.
- 4 Though rain falls throughout the year, there is a distinct summer maximum.
- 5 Of the annual precipitation of 76 to 152 cm, two-thirds come in the summer.
- 6 Winter is dry and cold, because the winds are dry westerlies that blow out from the continental interiors.

8 THE LAURENTIAN TYPE NATURAL VEGETATION



- 1 The predominant vegetation of the Laurentian type of climate is cool temperate forest.
- 2 The heavy rainfall, the warm summers and the damp air from fogs, all favor the growth of trees.
- 3 Generally speaking, the forest tend to be coniferous north of the 50° N parallel of latitude.

8 THE LAURENTIAN TYPE NATURAL VEGETATION

4 The increase in length and severity of the winter excludes forests that are not adaptable to cold conditions.

5 In the Asiatic region (eastern Siberia and Korea), the coniferous forest are, in fact, a continuation of the great coniferous belt of the taiga.

6 South of latitude 50°N , the coniferous forests give way to deciduous forests.

7 The occurrence of trees in almost pure stands, and the predominance of only a handful of species greatly enhance the commercial value of these forests.

8 As a result, they have been extensively felled for the extraction of temperate hardwood.

9 In many parts of Manchuria, Korea and Japan the forests have made way for the plough. Both food and cash crops are raised.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT



- 1 Lumbering and its associated timber, paper and pulp industries are the most important economic undertaking.
- 2 Agriculture is less important in view of the severity of the winter and its long duration.
- 3 Fortunately the maritime influence and the heavy rainfall enable some hardy crops to be raised for local needs.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

- 4 Hardy cereals like **oats** and **barley** can be sown and successfully harvested **before the onset of the cold winter.**



OATS



BARLEY

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT



SOYABEAN & SESAME

- 5 A number of other interesting crops are produced in the Asiatic region such as soya beans (northern China, Manchuria and Korea are amongst the world's leading producers), groundnuts, sesame, rape seeds and mulberry.
- 6 In the north American region, arable farming is not carried out on a sizable scale, except in the more favored localities.

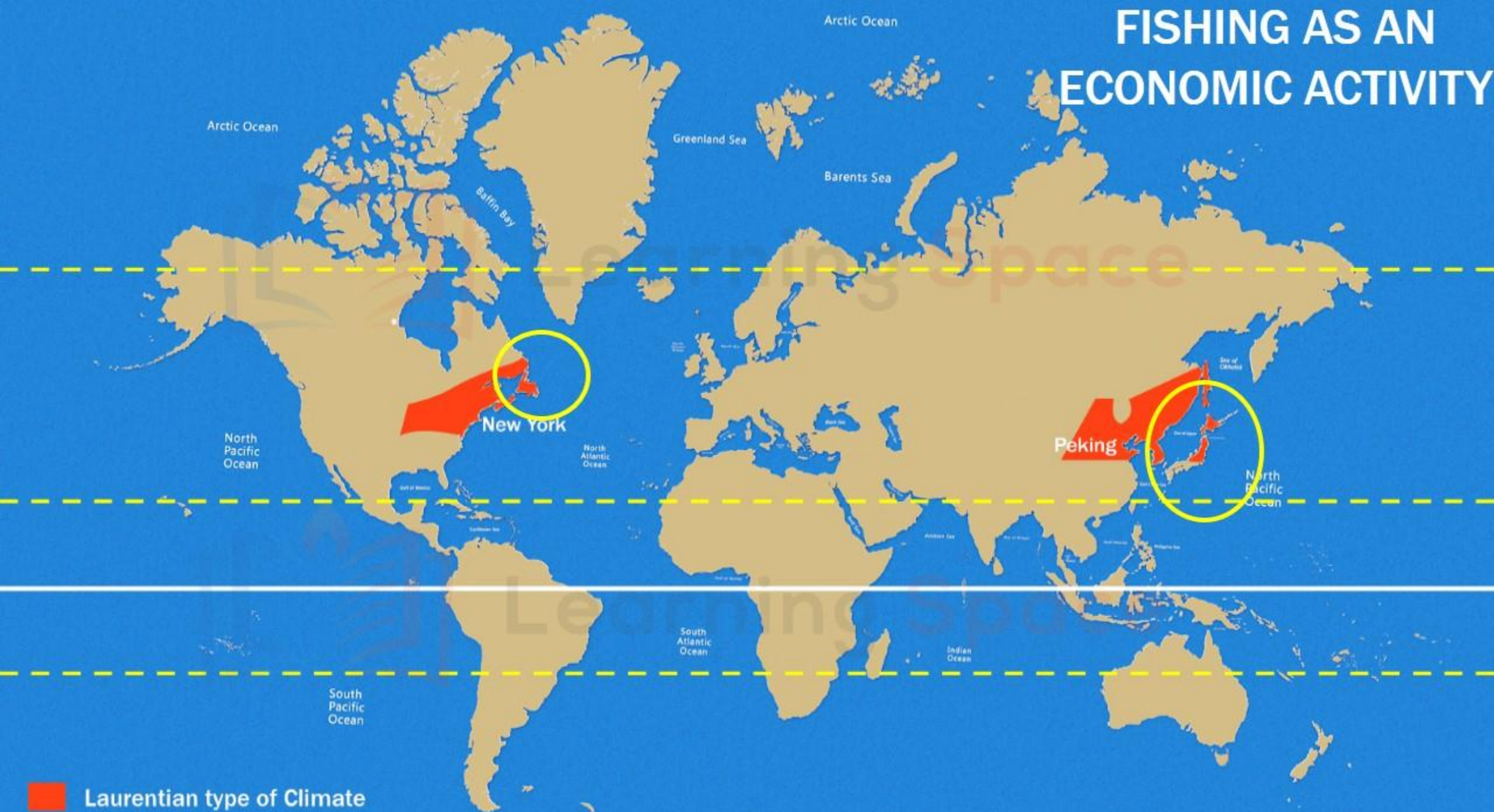
8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT



7 Farmers are engaged in dairy farming, hay cultivation and in mild maritime areas, fruit growing.

8 The fertile Annapolis valley in Nova Scotia is the world's most renowned region for apples. Fishing is, however, the most outstanding economic activity of the Laurentian climatic regions.

FISHING AS AN ECONOMIC ACTIVITY



ECONOMIC GEOGRAPHY

GS PAPER 1 QUESTION

POSSIBLE MAINS QUESTION

- 1** How is fishing said to be a predominant economic activity in the coastal lands of Japan?
(10 Marks) (200 words)
- 2** Geographically, explain how Japan and Newfoundland are good breeding grounds of fishes?
(12 Marks) (250 words)

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

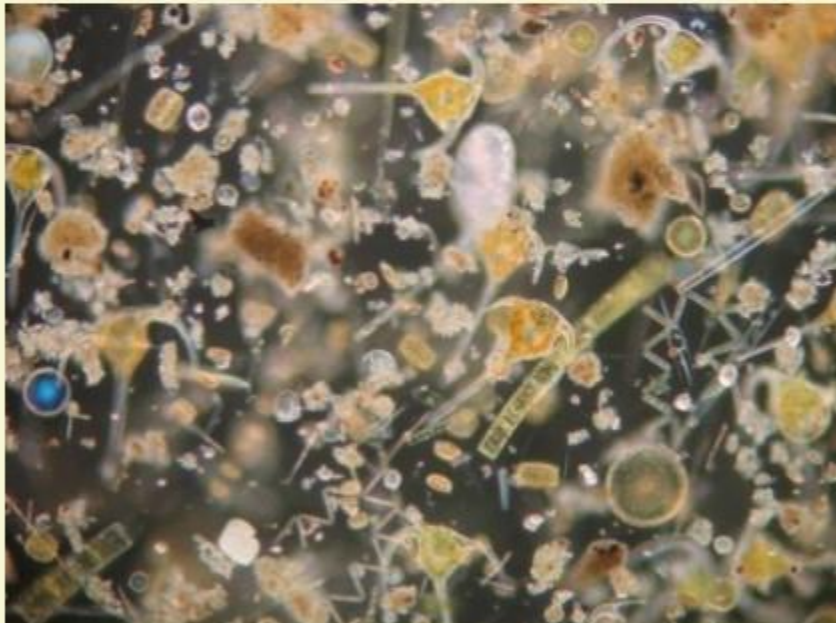
1. FISHING OFF NEWFOUNDLAND



- 1 This is one of the world's largest fishing grounds particularly on the Grand Banks of Newfoundland.
- 2 The reasons for its importance are geographical.
- 3 Fish feed on minute marine organisms, collectively called Plankton, which is present in abundance only in shallow waters adjacent to land masses, where sunlight can penetrate through.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

1. FISHING OFF NEWFOUNDLAND



4 The gently sloping continental shelves (less than 600 feet deep) which stretch for over 200 miles south-east of Newfoundland and off the coasts of the Maritime Provinces and New England contain a rich collection of microscopic plankton.

5 Fish of all types and sizes feed and breed here and support a thriving fishing industry not only in Canada and USA, but also in countries like Norway, France, Britain, Portugal, Denmark, Russia and Japan, who send fishing fleets to the Grand Banks.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

1. FISHING OFF NEWFOUNDLAND

6 As less than 1 percent of Newfoundland is cultivated, fishing provides employment for almost the entire population.

7 It is not only the chief source of wealth to Newfoundland, but also a major export item.

8 Both **pelagic fish** which live near the surface and **demersal fish**, which live near the bottom of shallow seas are caught.

9 The Chief fish caught is **cod** which is consumed fresh or dried, salted, smoked, canned or packed in ice for export to mainland America, Central and South America and Southern Europe. Cod liver oil is exported too.

8 THE LAURENTIAN TYPE

ECONOMIC DEVELOPMENT

1. FISHING OFF NEWFOUNDLAND



10 Further inland, in lakes and rivers, such as the St. Lawrence and the Great Lakes, freshwater fish, eg. Salmon, trout and eels are caught.

11 Fishing in this part of the world is highly specialized and very efficient.

12 In fact over-fishing is a growing problem. The United States Government and international fishing bodies are now contemplating strict measures in fish conservation.

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8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN



- 1 In the north-west Pacific, surrounding the islands of Japan, is another major fishing area of the world.
- 2 Nowhere else in the world are there so many people engaged in fishing as in this part of the north-west Pacific.
- 3 The mountainous nature of Japan and parts of mainland eastern Asia have driven many to seek a livelihood from the sea.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN



- 4 The scarcity of meat (there is little pasture in Japan for livestock farming of any kind) and religious reasons have popularised fish as the principal item of diet and the chief protein food of the Japanese and the Chinese as well.
- 5 Large quantities of fish and fish products are either canned or preserved for export to neighbouring countries.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN



Nori is a seaweed which is used in the making of the traditional Japanese cuisine- Sushi!

6 The Japanese also make use of fish wastes, fish meal and seaweeds as fertilizers in their farms.

7 Japan is one of the few countries that has taken to seaweed cultivation.

8 Coastal farms that are submerged in water grow weeds for sale as fertilizers, chemical ingredient and even as food.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

Another interesting aspect of Japanese fishing is pearl culture.



9 The divers of southern Japan dive down into the coastal waters and bring to the surface shell-fish called pearl oysters and extract the highly prized pearls for sale as ornaments.

10 As natural pearls in oysters are difficult to obtain in large numbers, the Japanese have begun to breed the young oysters.

11 By injecting tiny “seeds” into them, the oysters are made to secrete the pearl material, which accumulates to form cultured pearls. These are collected and exported.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN



- 12** Large whaling fleets complete with processing plants and experienced crews stay out in the open seas and return only occasionally for refueling or replenishment of fresh provisions.
- 13** As a nation, Japan accounts for a sixth of the world's total annual fish caught. She is the world's greatest fishing nation today

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 1- 80% NON-AGRICULTURAL LAND



- 1 Japan is not well endowed with natural resources, for as such as 80 percent of her land is classified Non-Agricultural. She has to take to the sea if she wants to survive.
- 2 This has compelled the people to develop the seas, and fishing has for centuries been the traditional occupation of many coastal Japanese.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 2- MEETING GROUND OF WARM AND COLD CURRENTS



The continental shelves around the islands of Japan are rich in plankton, due to the meeting of the warm Kuroshio and the cold Oyashio currents and provide excellent breeding grounds for all kinds of fish including herring, cod, mackerel, bonito, salmon, sardine and tuna, as well as crabs and lobsters.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 3- FAVOURABLE COASTLINE FOR PORTS



The indented coastline of Japan, provides sheltered fishing ports, calm waters and safe landing places, ideal for the fishing industry.

8**THE LAURENTIAN TYPE****ECONOMIC DEVELOPMENT****2. FISHING OFF JAPAN**

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 4- LESS CATTLE/POULTRY

1 Lack of lowlands and pastures means that only a few animals can be kept to supply meat and other protein food.

2 Fish in all its varied forms, fresh, canned, dried, frozen and in the form of fish pastes, fish sauce and spiced condiments takes the place of meat as Japan's primary source of protein food.

3 There is a great demand for it locally, and for export to other east Asiatic neighbours which lack the techniques of large scale commercial fishing.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 5- TECHNOLOGICAL PROGRESS



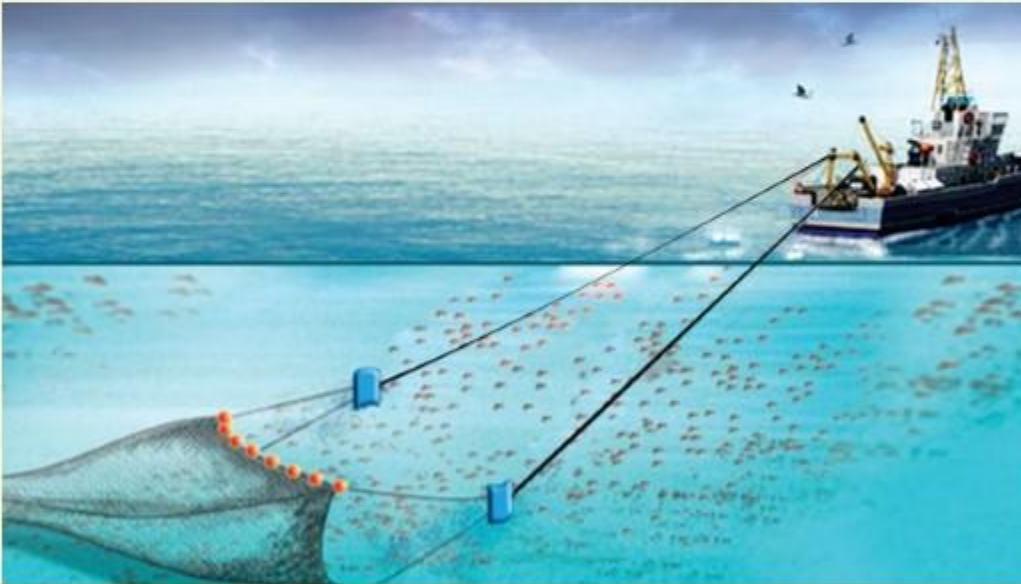
- 1 The Japanese fishermen began with small fishing boats, using nets, traps and lines.
- 2 With the progress made in industries, fishing has also become more scientific, aiming at heavy hauls, high returns and economy of time, effort and money.
- 3 Though three quarters of fishermen practise off-shore pelagic fishing either full-time or part-time in small boats, most of the commercial deep-sea demersal fishing is now highly mechanized.

8 THE LAURENTIAN TYPE ECONOMIC DEVELOPMENT

2. FISHING OFF JAPAN

LET US FIND OUT HOW THIS IS POSSIBLE...

REASON 5- TECHNOLOGICAL PROGRESS



- 4 Powered trawlers and modern refrigeration plants backed by sound financial organisations have greatly increased the annual fish yield.
- 5 Japan is now not only a major producer and exporter of fish and marine products, but also a centre for marine and fishing research.

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